

KRISHICLEAR

Final Competitive Strategy + Integrated Feature Blueprint

THE POSITION: Do not build another farmer marketplace. Build the clearing layer that decides which pooled supply should satisfy which pooled demand, at what farmer-safe price, with what delivered cost, and what happens when the plan fails.

PROBLEM

Multiple intermediary layers can reduce farmer realization while keeping consumer/bulk-buyer prices high.

PUBLIC COMPETITION

Publicly described PS-26033 concepts already cover marketplaces, forecasting, logistics, bidding, verification, contracts, aggregation and route optimization.

OUR CORE OBJECT

A temporary **Clearing Batch** — not a listing or ordinary order.

OUR DEFENSIBLE COMBINATION

Pooled demand + pooled supply + price corridor + route-aware clearing + ₹100 value ledger + automatic re-clearing.

01 Executive Position

The official PS asks for a digital marketplace connecting farmers/FPOs to consumers and bulk buyers, with logistics support and AI for demand forecasting and route optimization. That direction is now crowded: publicly visible concepts already describe direct marketplaces, demand forecasting, logistics/route optimization, live telemetry, bidding, contracts, verification and price intelligence. ■cite■turn0search0■turn0search3■turn0search6■

THE STRATEGIC SHIFT: Marketplace is the interface. Market clearing is the product.

The earlier research document found the strongest defensible gap to be an algorithmic clearinghouse: pooled buyer demand and farmer supply, a farmer-floor / buyer-ceiling corridor, route-aware economics, a transparent value ledger and native supplier-dropout re-clearing. ■filecite■turn2file0■L19-L29■

The uploaded KisanSetu blueprint adds strong farmer UX, opportunity discovery, net-earning comparison, voice/low-literacy access, supply-gap alerts, waste rescue and impact measurement. Those features are integrated below — but subordinated to the clearing engine so the product does not become another feature-heavy marketplace. ■filecite■turn3file0■L10-L54■

02 What Is Already Covered?

Capability	Publicly described in PS-26033 space	KrishiClear treatment
Direct farmer → buyer marketplace	Yes	Interface only; not the innovation
AI demand forecasting	Yes	Supporting signal, not the core decision
Route optimization	Yes	Integrated into clearing economics
Collective demand aggregation	Yes in strategy variants	Used as an input to clearing
Reverse logistics	Yes in strategy material	Priced jointly with the trade
Bidding / negotiation	Yes	Replaced by algorithmic clearing where feasible
Verification / trust	Yes	Quality evidence + auditable batch
Contracts / escrow	Yes	Settlement audit; not claimed as novelty
Price transparency	Yes in variants	Formal price corridor + ■100 ledger
Dropout recovery	Not prominently described in searched concepts	Native substitute-supply re-clearing
Core transaction object	Listing / order	Clearing Batch

03 Competitive Analysis — Publicly Described Concepts

This comparison is deliberately conservative: it compares what is publicly described, not private/unpublished team ideas. The public field is already crowded enough that “marketplace + AI + logistics” is not a sufficient differentiation claim. SIH Buddy explicitly warns that the problem is heavily cloned; public team material also demonstrates mature marketplace feature sets. ■cite■turn0search3■turn0search6■

Concept / source	Strongest visible features	Where it overlaps	What KrishiClear changes
KhetKonnnect	Farmer/FPO → consumer/bulk buyer; AI demand forecast; logistics/route optimization; price transparency	Marketplace + forecast + route	Clearing Batch and constrained allocation become the core.
E-Uzhavar Sandhai	Separate stakeholder portals; live stock/fleet telemetry; farm-gate pricing	Marketplace + operations + telemetry	Adds an explicit economic clearing decision across pooled supply/demand.
Crop Connect	Direct farmer-buyer flow: connect → match → deliver → settle	Direct matching + delivery	Moves from bilateral matching to many-to-many clearing.
AgriConnect	Real-time requests, bidding, negotiation, contracts, logistics, price intelligence, verification, AI modules	Marketplace + bidding + trust + AI	Algorithmic clearing reduces repeated bilateral negotiation and makes value split explicit.
SIH ONE strategy	Collective demand aggregation, reverse logistics, transparent escrow	Pooling + routing + settlement	Adds a formal price corridor, clearing objective and failure re-clearing.
KisanSetu blueprint	AI sell window, net earning simulator, farmer pooling, opportunity radar, digital mandi twin, waste rescue, impact ledger	Decision support + aggregation + logistics	These become UX/impact layers around the clearing engine, not a competing marketplace.

KEY FINDING: Almost every competitor answers “Where can I sell/buy?”
KrishiClear must answer “Which combination of supply and demand should be cleared together, under which constraints, at what economics, and how do we recover if the plan breaks?”

04 Complete Integrated Feature List

The following is the final product feature universe. The **Core** features define the novelty; **Support** features make the system deployable; **Wow** features make the demo memorable. This prevents overbuilding while retaining the full product vision.

Module	Feature	What it does	Priority
Demand	Buyer Demand Pool	Combines compatible RFQs by crop, grade, quantity, geography and time window.	CORE
Demand	Recurring Demand	Stores repeat institutional requirements and expected windows.	SUPPORT
Demand	Demand Confidence	Separates firm commitments from probabilistic forecast demand.	CORE
Supply	Farmer/FPO Supply Pool	Aggregates small lots by compatibility, harvest window and location.	CORE
Supply	One-Tap Sell Lot	Fast lot creation from crop, quantity, quality, harvest and pickup preference.	SUPPORT
Supply	Nearby Farmer Pool	Detects nearby compatible lots that unlock better fulfilment economics.	CORE
Pricing	Farmer-Floor / Buyer-Ceiling Corridor	Prevents the engine from proposing a trade below farmer minimum or above buyer ceiling.	CORE
Pricing	Fair-Price Explanation	Explains the effect of grade, distance, loss risk, demand pressure and logistics.	CORE
Pricing	Net Earning Simulator	Shows expected net ■ after transport, handling and other costs.	WOW
Clearing	Algorithmic Clearing	Solves constrained many-to-many allocation instead of simple one-to-one matching.	CORE
Clearing	Clearing Batch	Creates the auditable temporary trade object containing demand, supply, constraints, route and value split.	CORE
Clearing	Allocation Explainability	Shows why suppliers/lots were selected or rejected.	CORE
Logistics	Route-Aware Pricing	Includes route economics before trade confirmation.	CORE
Logistics	Multi-Pickup Route	Plans capacity-aware pickup and delivery.	CORE
Logistics	Reverse Logistics	Uses compatible return movement where it improves economics.	SUPPORT
Resilience	Substitute-Supply Engine	Replaces a failed commitment and re-clears the affected batch.	CORE
Resilience	Supply Gap Alert	Shows confirmed quantity versus remaining requirement and nearby sourcing options.	WOW
Resilience	Waste Rescue Mode	Reroutes cancelled/uneconomic produce to another buyer, processor or approved channel.	WOW
Trust	Quality Evidence Chain	Lot passport, grade, weight, timestamp, photos and inspection evidence.	CORE
Trust	Farmer Choice Guardrail	Farmer can set minimum realization and accept/reject a proposed clearing.	CORE
Trust	Verified Roles	Farmer/FPO/buyer/transporter verification and role-based access.	SUPPORT
Economics	■100 Margin Leak Ledger	Shows farmer payout, logistics, storage, platform fee and approved costs.	CORE

Module	Feature	What it does	Priority
Economics	Impact Ledger	Farmer uplift, buyer saving, km optimized, consolidation and estimated waste avoided.	WOW
AI	Demand Forecast	Forecasts demand by crop/zone/time with confidence.	SUPPORT
AI	Opportunity Radar	Maps demand pockets, price opportunity, available supply and logistics feasibility.	WOW
AI	Match Ranking	Ranks feasible lots using reliability, grade, timing, distance and logistics cost.	CORE
AI	Exception Prediction	Predicts under-fulfilment risk and recommends substitutes.	CORE
UX	Voice / Low-Literacy Mode	Local-language labels, voice-assisted lot creation and spoken recommendations.	SUPPORT
UX	Offline / Low-Bandwidth FPO Mode	Batch upload and later confirmation for weak-connectivity environments.	SUPPORT
Operations	Digital Mandi Twin	Interactive supply, arrivals, demand and price-pressure map.	WOW
Operations	Shipment Timeline	Pickup proof, delivery proof, ETA and exception states.	SUPPORT
Payments	Transparent Settlement	Payment intent, invoice, settlement and refund status.	SUPPORT
Governance	Audit Log	Decision, change and settlement history for traceability.	CORE
AI Governance	Confidence + Model Version	Shows data freshness, model version, confidence and fallback rules.	SUPPORT

05 The Five Features That Actually Make You Different

1 • CLEARING BATCH

The core object is a temporary algorithmically formed trade: pooled demand + pooled supply + quality constraints + pickup/delivery nodes + price corridor + auditable value split. ■filecite■turn2file6■L292-L305■

2 • ROUTE-AWARE CLEARING

A trade is not accepted merely because price matches. Logistics cost is part of the clearing decision, so a “cheap” farm that is expensive to collect can lose to a closer feasible combination.

3 • ■100 MARGIN LEAK LEDGER

Every buyer ■100 is decomposed into farmer realization, logistics, storage, platform and approved costs. This turns “remove intermediaries” into a measurable claim. ■filecite■turn2file6■L300-L307■

4 • AUTOMATIC RE-CLEARING

If one farmer fails, the platform does not just show an error. It searches compatible substitutes and recomputes allocation + route. The research scan found this was not prominently described in the searched PS-26033 concepts. ■filecite■turn3file5■L214-L224■

5 • FARMER-SAFE PRICE CORRIDOR

The system respects a farmer minimum realization and buyer maximum delivered price, rather than optimizing a single marketplace commission or raw selling price.

ONE-LINE DIFFERENTIATOR:

“Others connect buyers and farmers. KrishiClear computationally clears the trade between pooled buyers and pooled farmers — including price, logistics, transparency and failure recovery.”

Do not claim legal/patent uniqueness. The correct claim is: **this exact combined mechanism was not found in the public PS-26033 material searched.** The uploaded research explicitly makes this distinction.

■filecite■turn2file0■L19-L29■

06 Farmer Experience — Integrated from the KisanSetu Blueprint

The uploaded KisanSetu blueprint has several excellent usability ideas. They should remain, but their output must feed the clearing engine rather than becoming a separate marketplace experience. ■filecite■turn3file2■L91-L131■

Farmer screen	Final KrishiClear behaviour	Why it matters
Farmer Command Center	"What can I sell? Where? For how much? What should I do next?"	Turns complexity into action.
AI Sell Window	Shows demand signal, price range, confidence and reason.	AI becomes an actionable recommendation, not a dashboard.
Net Earning Simulator	Compares feasible buyer/market routes after logistics and costs.	Optimizes net realization, not gross price.
One-Tap Sell Lot	Creates a structured supply commitment.	Feeds the supply pool immediately.
Price Shield	Current/reference/predicted range + uncertainty.	Prevents false certainty around AI price forecasts.
Nearby Farmer Pool	Suggests consolidation when volume unlocks a better clearing.	Connects smallholders to pooled demand.
Voice / Low Literacy	Voice + local language + large actions.	Makes the system realistic beyond a judging laptop.

The KisanSetu “Opportunity Radar”, “■ Before vs ■ After”, “One Buyer Order → Many Farms”, “Digital Mandi Twin”, “Supply Gap Alert”, “Waste Rescue Mode” and “Impact Ledger” are retained as high-impact presentation layers. ■filecite■turn3file0■L14-L54■

07 Buyer, FPO & Operations Experience

BUYER CONSOLE Create one requirement or recurring RFQ. Set quantity, grade, delivery window and maximum delivered price.	POOLED DEMAND VIEW See how individual buyer requirements merge into demand pools and how much is firm vs forecast.
CLEARING PREVIEW Before accepting, show proposed farmer lots, quantity, route, delivered cost and value split.	FPO COMMAND CENTER Upload multiple farmers, confirm quantities, quality, harvest window and minimum realization.
LOGISTICS CONSOLE See vehicle capacity, pickup sequence, ETA, cost/km and empty-km reduction.	QUALITY / LOT PASSPORT One lot carries grade evidence, weight, timestamps, inspection and exception history.
SETTLEMENT CENTER Show payment status, farmer payout, logistics and other approved costs.	ADMIN / GOVERNANCE Manage roles, exceptions, audit trail, model versions, configuration and data quality.

The uploaded endpoint blueprint already maps these domains across auth/roles, farms/lots, RFQs, matching, pricing/AI, orders, consolidation, logistics, tracking, payments, quality, disputes, GIS, analytics, admin and AI model operations.

filecite turn3file3 L142-L167

08 How the Clearing Algorithm Works

Stage	Inputs	Output
1. Demand normalization	Crop, grade, quantity, delivery time, geography, buyer ceiling	Standard demand objects
2. Supply normalization	Quantity, grade, harvest window, geography, farmer floor	Standard supply objects
3. Candidate generation	Quality + geography + time compatibility	Feasible farmer–buyer edges
4. Price corridor	Reference price + grade + distance + expected loss + demand pressure	Farmer floor / buyer ceiling
5. Match scoring	Reliability + distance + grade + timing + expected logistics	Ranked feasible edges
6. Allocation	Quantities + constraints + objective function	Winning allocations / Clearing Batches
7. Route optimization	Pickup/drop nodes + vehicle capacity + time windows	Least-cost feasible route
8. Settlement audit	Final buyer payment + approved costs	Farmer payout + ledger
9. Failure simulation	Supplier dropout / delay / shortage	Replacement + re-cleared route

Judge-friendly objective: maximize farmer payout + fulfilled demand value – logistics cost – spoilage penalty – unmet-demand penalty, subject to quantity, quality, harvest-window, route-capacity and farmer-minimum constraints. The earlier research recommends OR-Tools for routing/optimization, Python/FastAPI for the service and PostgreSQL/PostGIS for geospatial data.

■filecite■turn2file6■L308-L319■

Important positioning: AI predicts demand/risk/price signals; the deterministic optimizer performs constrained clearing. Do not label a route solver itself as “AI”. ■filecite■turn3file5■L205-L213■

09 Feature-by-Feature Competitive Matrix

Legend: ✓ = publicly described; ■ = partial/adjacent; — = not prominently described in the public material reviewed. This is a public-evidence matrix, not a claim about private implementations.

Capability	KhetKonne ct	E-Uzhavar	Crop Connect	AgriConne ct	SIH ONE	KisanSetu	KrishiClear
Direct marketplace	✓	✓	✓	✓	✓	✓	Interface only
Demand forecasting	✓	■	—	✓	■	✓	✓ supporting signal
Route optimization	✓	✓	■	✓	✓	✓	✓ integrated into clearing
Collective demand	■	■	■	■	✓	✓	✓ core input
Bidding / negotiation	—	■	■	✓	—	✓	Replaced by clearing where feasible
Price transparency	✓	✓	■	✓	✓	✓	Formal corridor + ■100 ledger
Quality / verification	■	✓	■	✓	■	✓	Lot evidence chain
Farmer net-earning simulation	—	—	—	■	—	✓	✓
Opportunity radar / mandi twin	—	■	—	■	—	✓	✓ optional WOW
Waste rescue	—	—	—	■	—	✓	✓
Clearing Batch object	—	—	—	—	—	—	✓
Farmer floor + buyer ceiling	—	—	—	—	—	■	✓
Route-aware price clearing	—	—	—	—	■	■	✓
Automatic dropout re-clearing	—	—	—	—	—	■	✓
Margin Leak Ledger	—	—	—	—	■	✓ impact ledger	✓ core
Core success metric = farmer realization + delivered price + leakage	—	—	—	—	—	■	✓

Public descriptions supporting this matrix include the AgriConnect repository and the uploaded competitive scan. ■cite■turn0search6■
 ■filecite■turn3file7■L306-L322■

10 The “Judge-Wow” Demo — One Story, Not 30 Features

The uploaded KisanSetu blueprint recommends a single connected 5-minute scenario rather than disconnected features. We strengthen that flow by making the clearing engine the hero. ■filecite■turn3file4■L177-L194■

Time	What happens	What proves uniqueness
0:00–0:30	Show conventional chain and farmer/consumer price gap.	Problem framing + baseline.
0:30–1:00	Farmer creates a lot; AI Sell Window shows opportunity.	Actionable AI.
1:00–1:40	Buyer creates 5,000 kg demand.	Demand pooling starts.
1:40–2:20	System combines 3 farmers + 1 FPO and creates a Clearing Batch.	Many-to-many clearing.
2:20–3:00	Show price corridor and ■100 ledger.	Transparent economics.
3:00–3:40	Route engine compares naive vs cleared route.	Route-aware economics.
3:40–4:15	Cancel one farmer.	Native substitute re-clearing.
4:15–4:40	Show Supply Gap Alert / Waste Rescue if needed.	Resilience.
4:40–5:00	Impact dashboard: farmer uplift, buyer saving, km, fulfilment, waste.	Proof, not marketing.

THE MOMENT TO MAKE MEMORABLE:

“A farmer did not get a better price because we made a prettier marketplace. The system mathematically found a better feasible trade for the entire batch — and when one supplier failed, it solved the trade again.”

11 Metrics That Prove the Solution

Area	Metric	Definition / evidence
Farmer economics	Farmer realization share	Farmer payout ÷ buyer payment.
Farmer economics	Farmer price uplift	KrishiClear realization vs selected baseline/reference.
Buyer economics	Delivered price delta	Baseline estimated delivered price – KrishiClear delivered price.
Leakage	Intermediary leakage	Buyer payment – farmer payout – auditable logistics/storage/platform costs.
Fulfilment	Fulfilment rate	Delivered quantity ÷ committed quantity.
Logistics	Empty km	Distance travelled without productive pickup/delivery.
Logistics	km/kg	Route distance ÷ delivered quantity.
Resilience	Re-clearing time	Seconds/minutes to replace a failed supplier and recompute route.
Waste	Spoilage avoided	Estimated baseline loss – cleared-batch loss.
AI	Forecast error	Hold-out validation metric; show confidence/calibration.
Trust	Quality dispute rate	Disputed lots ÷ completed lots.
Impact	Consolidation ratio	Cleared quantity achieved through pooled supply vs fragmented dispatch.

The earlier research and KisanSetu blueprint both emphasize outcome metrics rather than feature count, including farmer realization, buyer savings, fulfilment, logistics efficiency, forecast quality, trust and impact.

filecite turn2file4 L228-L235 filecite turn3file1 L65-L80

12 Final Technical Architecture

Layer	Recommended component	Role
Farmer / FPO PWA	React / Flutter + offline cache	Lot creation, commitments, voice/local-language UX.
Buyer web app	React / Next.js	RFQs, demand pools, clearing preview, settlement.
Operations console	Web dashboard + GIS	Clearing batches, routes, exceptions, quality and impact.
API	FastAPI / Node	Business rules and orchestration.
Clearing service	Python + OR-Tools / MILP	Allocation, corridor constraints and re-clearing.
ML service	Python	Demand, fair-price, reliability and exception models.
Data	PostgreSQL + PostGIS	Lots, demand, batches, routes and spatial queries.
Queue	Redis / task queue	Forecasting, clearing and route jobs.
Maps	GIS provider	Pickup/drop visualization and route geometry.
Audit	Append-only event log pattern	Decision and settlement traceability.
Model ops	Versioning + confidence + drift checks	Responsible AI and reproducibility.

Architecture principle: every AI/optimization output must end in a user action or measurable KPI. Avoid adding chatbot, blockchain, AR/VR or unrelated modules merely for novelty; current PS guidance also warns against over-scoping and recommends one deeply validated scenario. ■[cite](#)■[turn0search7](#)■

13 What to Actually Build for SIH

DO NOT BUILD THE WHOLE PLATFORM.
Build one commodity + one geography + one complete clearing scenario, then make the engine visibly excellent.

MVP module	Build now	Demo evidence
Buyer console	Create 3–8 demand requests	They merge into one demand pool.
Farmer/FPO console	Create 15–30 synthetic supply lots	Small lots become a usable supply pool.
Clearing engine	Price corridor + allocation	Winning Clearing Batch in seconds.
100 ledger	Farmer/transport/storage/platform split	Judge sees where money goes.
Route engine	Multi-pickup route	Map + cost comparison.
Exception engine	Drop one supplier	Automatic replacement + recalculated route.
Farmer UX	Sell window + net earning option	Actionable recommendation.
Impact dashboard	Before/after KPIs	Evidence of farmer/buyer/logistics improvement.

This is aligned with the earlier research MVP recommendation: buyer console, FPO/farmer console, clearing engine, price ledger, route engine, exception engine and judge dashboard. [filecite turn2file5 L257-L273](#)

14 Judge Questions — Final Answers

“ISN’T THIS JUST ANOTHER MARKETPLACE?”

No. Marketplace is only the input/output interface. The core is constrained market clearing across pooled demand and pooled supply.

“BUT COLLECTIVE DEMAND ALREADY EXISTS.”

Yes. We are not claiming aggregation itself is new. Our differentiation is what happens after aggregation: corridor-constrained, route-aware clearing with an auditable value split.

“ISN’T THIS JUST eNAM?”

Do not claim to replace eNAM. Position KrishiClear as a transaction-clearing and logistics layer that can use existing market channels as sources/sinks.

■filecite■turn3file5■L205-L213■

“WHERE IS THE AI?”

Demand, fair-price, reliability and exception models provide predictive signals. Optimization performs the actual constrained decision. This is a stronger AI story than calling routing alone AI.

“WHY WILL FARMERS TRUST IT?”

The farmer sets a minimum realization, sees the proposed economics before accepting and receives an auditable payout/cost breakdown.

“WHAT IF A FARMER DROPS OUT?”

The affected Clearing Batch is re-solved with compatible substitute supply; the route is recomputed.

“WHAT IF THE FORECAST IS WRONG?”

Forecast demand is explicitly probabilistic. Firm commitments and predicted demand are separated, confidence is shown, and fallback rules are defined.

“WHAT MAKES IT MEASURABLE?”

Farmer realization share, delivered price, leakage, fulfillment, km/kg, re-clearing time and spoilage avoided are direct outcome metrics.

15 Final Competitive Position

After integrating the previous KrishiClear research with the uploaded KisanSetu feature blueprint and the current public PS-26033 landscape, the recommended product hierarchy is:

LEVEL 1 — CORE NOVELTY

Algorithmic clearinghouse + Clearing Batch + farmer-floor/buyer-ceiling corridor.

LEVEL 2 — ECONOMIC PROOF

■100 Margin Leak Ledger + farmer realization + buyer delivered cost.

LEVEL 3 — PHYSICAL REALITY

Route-aware clearing + multi-pickup + reverse logistics + substitute re-clearing.

LEVEL 4 — INTELLIGENCE

Demand forecast + fair-price model + reliability + exception prediction.

LEVEL 5 — HUMAN USABILITY

Farmer Command Center + net earning simulator + voice/low-bandwidth + local language.

LEVEL 6 — WOW / IMPACT

Opportunity Radar + Digital Mandi Twin + Supply Gap Alert + Waste Rescue + Impact Ledger.

THE FINAL PITCH:

“Existing solutions help farmers and buyers find each other. KrishiClear goes one layer deeper: it computes the trade itself. It pools fragmented demand and supply, protects a farmer minimum realization, accounts for logistics before clearing, exposes every ■100, and automatically re-clears the batch when reality changes.”

This is the position most defensible from the public evidence reviewed. It is not a legal guarantee of novelty or proof that an unpublished SIH team has not independently conceived the same mechanism. ■filecite■turn2file0■L24-L29■

16 Research Sources & Evidence Base

Public web research was performed on 12 September 2026. The official/community archive confirms the PS asks for a direct marketplace, logistics support, demand forecasting and route optimization. ■cite■turn0search0■

Source	Use in this document
SIH 2026 PS 26033 public archive	Problem statement and expected solution.
SIH Buddy — PS 26033	Crowding / cloning assessment and public competitive framing.
SIH ONE — PS 26033	Collective demand, reverse logistics and escrow strategy reference.
AgriConnect public GitHub repository	Public evidence of marketplace, bidding, contracts, verification, AI and logistics features.
Uploaded: SIH_2026_PS_26033_Deep_Research_Uniq ue_Solution_v2.pdf	Prior competitive scan, KrishiClear concept, feature set, algorithm, KPIs and novelty caveat.
Uploaded: KisanSetu_Out_of_Box_Feature_Blueprint.pdf	Farmer UX, opportunity radar, net earning simulator, wow features, endpoint blueprint, demo and metrics.
Thirai PS-26033 page	Independent technical interpretation and MVP/scope guidance.

Important evidence note

The public sources do not constitute a complete registry of every SIH team's private/internal idea. Therefore, all novelty language in this report is intentionally scoped to **publicly discoverable PS-26033 concepts reviewed**.

Public references

<https://sih2026.vuce.in/ps/SIH26033>
<https://www.sihbuddy.in/ps/SIH26033>
<https://sihone.pages.dev/ps/26033>
<https://github.com/adanyanandini/nandini-sih-agriculture-demo>
<https://www.thirai.in/sih2026/sih26033>

Prepared for SIH 2026 • PS 26033 • KrishiClear • Final integrated competitive and feature document • 12 September 2026

KrishiClear — Team Additions Addendum

PS 26033 | Additive update to the existing competitive feature document

Important: The original KrishiClear document is not being replaced or re-written. These pages only add the team-suggested ideas, merge overlaps with the existing architecture, and rank them by impact × uniqueness × hackathon feasibility.

Keep unchanged	All earlier KrishiClear sections, competitive analysis, clearing-batch concept, price corridor, ₹100 ledger, automatic re-clearing, farmer guardrails, architecture, MVP and demo story.
Newly integrated	Freshness limit, commodity temperature profiles, route fuel economics, profit-maximizing routing, what-if simulation, additional simulation engines, APMC/legal compliance layer, AI forecasting sharpening, FPO analytics dashboard, multilingual UX.
Priority rule	Build only the P0 items deeply. P1 items should be working but simple. P2 items are demo/wow extensions only if time remains.

The uploaded team document already emphasizes net-earning calculation, AI price/demand prediction, smart aggregation, logistics/route planning, voice/low-data access, and a clear priority/demo order. The additions below strengthen those same components rather than creating a second unrelated product.

1. P0 — The Four Highest-Value Additions

These are the additions most worth building because they directly reinforce KrishiClear's central differentiator: **clearing a trade while respecting economics, time, quality and logistics together.**

1.1 Freshness Limit / Remaining Freshness Engine

What it does: Every Clearing Batch gets a freshness window. The system estimates how much usable selling time remains before quality risk becomes material.

MVP: Maintain a commodity profile containing expected shelf-life, handling sensitivity and an acceptable temperature band. Track harvest time → pickup → consolidation → transit → delivery. Output **Freshness Remaining %** and a simple risk label: Safe / Watch / Critical.

Why it is different: Route optimization is no longer only about distance. The optimizer is solving for **value before spoilage**.

1.2 Commodity-Specific Temperature Intelligence

Do **not** use one universal 'best temperature for vegetables'. Store a crop/commodity profile with its recommended temperature range, humidity sensitivity and temperature-exposure tolerance. The engine then compares the actual/assumed transit profile against that range.

Demo: Select Tomato → show its target band → increase route temperature → freshness score falls → expected realization falls → engine proposes an alternate route/vehicle.

1.3 Route Cost + Profit Maximization

Extend the existing route planner so that it calculates **delivered economics**, not merely shortest distance.

Cost inputs: distance, vehicle mileage, petrol/diesel cost, tolls, driver cost, loading/unloading, cold-chain/refrigeration cost where applicable, and vehicle capacity.

Optimization objective: maximize farmer net realization subject to buyer deadline, vehicle capacity, route time and freshness constraints. This plugs directly into the existing Net-Earning Calculator and Clearing Batch.

1.4 What-If Simulation Engine

Add a small simulation panel: **"What happens if conditions change?"** The user can change traffic, temperature, delay, fuel cost or vehicle availability and immediately see the effect on delivery time, freshness, logistics cost and farmer net earning.

Hackathon MVP: rule-based sensitivity calculations are sufficient. The important part is that the system shows a before/after decision and explains why the clearing should change.

2. P1 — Operational Intelligence & Compliance

These features should be implemented in a lightweight but credible way. They add institutional realism without distracting from the core clearing engine.

2.1 APMC / Mandi Legal Compliance Layer

For the Maharashtra pilot, the product should treat APMC compliance as a **decision/guardrail layer**, not as hard-coded legal advice. Maharashtra's current framework contains provisions covering direct marketing, private markets, farmer-consumer markets, electronic trading platforms, licences and dispute/payment mechanisms.

■cite■turn0search0■turn0search2■

Feature: Before a trade is confirmed, show: *Origin state* → *Destination state* → *Commodity* → *Transaction mode* → *Applicable compliance path* → *Required licence/partner status* → *Fee/tax/legal-review flag*.

Important: Because APMC rules and permissions are state-specific, the production version should use a configurable rules table and human/legal review for ambiguous cases. Maharashtra's official FAQ also distinguishes regulated market purchases from permitted direct-marketing/private-market channels and licensing requirements.

■cite■turn0search1■turn0search3■

2.2 AI Demand + Price Forecasting — Sharpened

Keep the existing AI forecasting feature, but make it **decision-oriented**. Forecast demand/price by commodity × region × date window, then feed the output into the Clearing Batch.

Output: expected demand range, confidence, top drivers, and recommended sell window. Example: "Demand likely HIGH in Pune in 2–4 days; current route economics support clearing now."

Data: historical prices/arrivals, seasonality, orders, current marketplace demand and—when available—weather signals. For the hackathon, a simple time-series/regression model is enough; the competitive value comes from connecting the forecast to an executable clearing decision.

2.3 FPO Command Center

A dedicated FPO dashboard should aggregate what individual farmer screens cannot show.

Show: **total orders, fulfilled quantity, farmer earnings, average realization/kg, logistics cost/kg, rejection/spoilage rate, average buyer price, pending payments, top commodities, top buyer clusters, and month-on-month trend.**

Unique extension: add a "*Money recovered by optimization*" card — estimated additional farmer realization generated by aggregation + route optimization + reduced spoilage compared with the baseline.

2.4 Multilingual + Voice Continuation

The team document already proposes voice listing and low-data/SMS access. Extend this to the complete farmer decision flow: listing, price explanation, buyer offer, route recommendation, confirmation and alerts.

Priority languages for the demo: English + Hindi + Marathi. Architect strings separately so more languages can be added later.

3. Simulation Engine Roadmap — The 'Digital Twin Lite'

Do not build five unrelated AI models. Build one reusable simulation framework with multiple scenario modules. This makes the feature look advanced while remaining achievable.

Simulation	Input changed	What changes	Decision produced
Traffic Shock	Traffic +20/40/60%	ETA, fuel cost, freshness	Reroute / re-clear / vehicle swap
Temperature Shock	Transit temperature above target	Freshness score, spoilage risk	Cold-chain route / faster route
Fuel Price Shock	Petrol/diesel █/L	Route cost, farmer net	Best economic route
Delay Shock	Pickup/handling delay	Remaining freshness	Change allocation / priority
Supplier Dropout	Farmer unavailable	Fulfilment gap	Automatic re-clearing
Demand Shock	Buyer demand ±%	Clearing quantity / price	Re-pool supply/demand

The strongest demo sequence

1) Create a Clearing Batch → 2) show best route and farmer net earning → 3) turn traffic up → 4) show freshness/cost impact → 5) engine proposes a different route → 6) show the revised clearing and explain the █ impact.

This is a stronger 'wow' moment than simply showing a forecast chart because the prediction **changes an actual trade decision**.

3.1 Reusable Simulation Architecture

Use one scenario object: **baseline state + changed variables + constraints + recalculation + comparison**. Keep the engine deterministic for the demo. Later, it can be connected to live traffic, weather and fuel-price feeds.

4. Final Priority Stack + 1% WOW Factor

The goal is not to add every possible feature. The goal is to make the existing KrishiClear concept visibly harder to copy while remaining buildable.

Priority	Feature	Build depth	Reason
P0	Freshness + commodity temperature profile	Working	Adds perishability intelligence to clearing
P0	Route fuel economics + profit optimizer	Working	Turns routing into farmer-profit optimization
P0	What-if simulation engine	Working with synthetic scenarios	Best visual judge moment
P1	APMC/legal compliance guardrail	Rules + flags	Institutional credibility
P1	FPO Command Center	Working analytics	Shows scale beyond one farmer
P1	AI demand/price forecasting	Simple explainable model	Supports sell-window decisions
P1	Multilingual + voice	3 languages	Real rural usability
P2	Extra simulation modules	Demo only	Wow without core-risk

4.1 The 1% WOW Feature — “Counterfactual Profit Proof”

After KrishiClear clears a batch, show a small comparison card: “**What would likely happen under the next-best alternative?**”

KrishiClear: Farmer realization ■X | Logistics ■Y | Freshness risk LOW | Delivery 16:20

Alternative: Estimated farmer realization ■X–■Δ | Logistics ■Y+■Δ | Freshness risk MEDIUM | Delivery 18:05

Then state the reason in one sentence: “**This clearing was selected because it preserves ■Δ farmer realization while staying inside the freshness and buyer-deadline constraints.**”

This is not another marketplace feature. It is an **auditable proof of why the engine made the decision.**

4.2 Other additions worth considering later

- **Quality Evidence Passport:** harvest time, lot weight, grade, temperature exposure and inspection events travel with the lot.
- **Spillage/Waste Rescue Mode:** if freshness risk becomes critical, automatically search for a closer buyer/processor instead of letting the batch fail.
- **Impact Ledger:** farmer ■ gained, kilometres avoided, spoilage avoided and buyer price reduction compared with baseline.
- **Auto-Reclear Trigger:** a supplier, route or buyer change automatically re-solves only the affected Clearing Batch.
- **Explainability trace:** every major clearing decision records the top constraints and economic reasons.

Final recommendation

Do not add more dashboards just for quantity. Add intelligence where one variable changes another: temperature → freshness → route → cost → farmer profit → clearing decision. That chain is the real differentiator.